Signature Verification using Convolutional Neural Networks and Feature Extraction Techniques (HOG & SIFT)

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***Abstract*—This study presents a comparative analysis of Convolutional Neural Networks (CNN) and traditional feature extraction techniques for signature verification. Specifically, Histogram of Oriented Gradients (HOG) and Scale-Invariant Feature Transform (SIFT) descriptors were extracted and classified using Support Vector Machines (SVM). The dataset consists of genuine and forged signatures from multiple users. Model performances were evaluated using accuracy, precision, recall, and F1-score metrics. Experimental results indicate that CNN and HOG-SVM models achieve high accuracy, while SIFT-SVM underperforms due to descriptor inconsistency and sparse keypoint detection.**

***Index Terms*—CNN, HOG, SIFT, Signature Verification, SVM, Feature Extraction, Image Processing**

I. Introduction

Signature verification is a critical task in document authentication, legal processes, and identity verification. Accurate classification of signatures ensures security and prevents forgery. This study investigates deep learning-based approaches using CNNs for automated feature learning and compares them with classical handcrafted feature extraction techniques such as HOG and SIFT combined with SVM classifiers. The primary objective is to determine which methodology yields the highest accuracy and reliability in recognizing signature patterns across multiple users.

II. Methodology

A. Dataset Description

The experiments utilize the publicly available Signature Verification Dataset [1], comprising genuine and forged signatures of multiple users. Images are organized into training and validation sets with user-specific folders, each containing multiple signature samples.

B. Preprocessing

All signature images were converted to grayscale to reduce computational complexity while preserving essential stroke information. Images were resized to 128×128 pixels and normalized to the range $[0, 1]$. For CNN training, user identities were label-encoded and converted to categorical format.For HOG and SIFT extraction, grayscale images were directly processed to obtain descriptors suitable for SVM classification.

C. CNN Architecture

The CNN model consists of three convolutional blocks with increasing filter sizes (32, 64, 128) and ReLU activation, followed by max-pooling layers. Flattened feature maps are fed to a dense layer of 256 units with dropout (0.5). The output layer uses Softmax activation for multi-class classification. The model was trained using Adam optimizer and categorical cross-entropy loss for 15 epochs and a batch size of 32.

TABLE I
CNN Architecture Summary

Layer Type	Output Shape
Conv2D (32 filters)	(126,126,32)
MaxPooling2D	(63,63,32)
Conv2D (64 filters)	(61,61,64)
MaxPooling2D	(30,30,64)
Conv2D (128 filters)	(28,28,128)
MaxPooling2D	(14,14,128)
Flatten	25088
Dense (256) + Dropout	256
Dense (Output)	64

D. HOG Feature Extraction

Histogram of Oriented Gradients (HOG) captures the distribution of intensity gradients and edge directions within localized image regions. The image is divided into cells where gradient orientations are computed. Blocks of cells are normalized to improve invariance against illumination and contrast variations.

In this study, HOG descriptors were computed using:

- 9 gradient orientations
- Cell size: 8×8
- Block size: 2×2

The extracted features were then used to train a linear SVM classifier which learns to distinguish between different users based on these gradient-based representations. The effectiveness of HOG lies in its ability to encode both local texture variations and global shape structures of signatures. [2] illustrates the HOG feature visualization of a sample signature, where the gradient orientations highlight the prominent strokes and curvature patterns, which are essential for discriminating genuine signatures from forgeries.

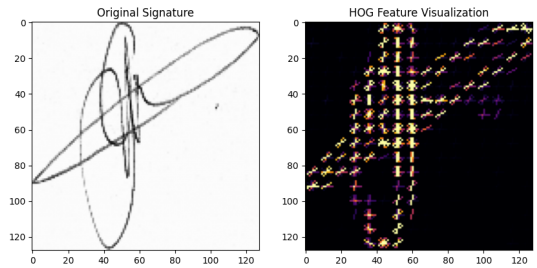

Fig. 1. Visualization of HOG features from a sample signature image.

E. SIFT Feature Extraction

Scale-Invariant Feature Transform (SIFT) detects distinctive keypoints in images that are invariant to scale and rotation. Each keypoint is represented by a 128-dimensional descriptor capturing the local gradient distribution. These descriptors were flattened to create fixed-length feature vectors before training an SVM classifier.SIFT is particularly effective for capturing fine-grained local patterns but may suffer when keypoint density varies across samples. [3] shows SIFT keypoints detected on a sample signature, demonstrating its ability to capture distinctive signature strokes.

Fig. 2. SIFT keypoint visualization on a sample signature.

III. Results

A. CNN Performance

The CNN achieved an accuracy of 92.7% on the validation set, with high precision and recall metrics, indicating effective feature learning and discrimination between users.

TABLE II
CNN Validation Metrics

Metric	Value
Accuracy	0.9269
Precision (macro)	0.9532
Recall (macro)	0.9297
F1 Score (macro)	0.9307

B. HOG + SVM Performance

HOG features combined with SVM achieved slightly higher validation accuracy (93.8%), demonstrating that handcrafted gradient-based features can remain competitive for structured datasets.

C. SIFT + SVM Performance

The SIFT-based model underperformed with 33.7% accuracy due to descriptor variability and sparse keypoint coverage.

TABLE III
HOG + SVM Validation Metrics

Metric	Value
Accuracy	0.9382
Precision (macro)	0.9531
Recall (macro)	0.9271
F1 Score (macro)	0.9240

TABLE IV
SIFT + SVM Validation Metrics

Metric	Value
Accuracy	0.3371
Precision (macro)	0.2901
Recall (macro)	0.2974
F1 Score (macro)	0.2725

IV. Training Curves

[4] illustrates CNN accuracy and loss over 15 epochs. The curves indicate stable convergence, minimal overfitting, and consistent validation performance.

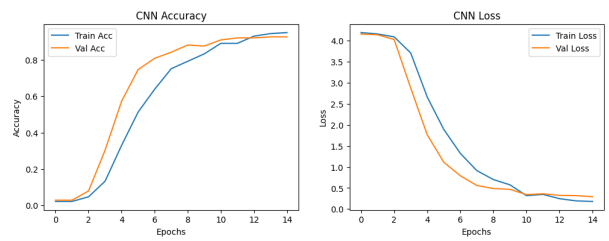

Fig. 3. CNN training and validation accuracy and loss curves over 15 epochs.

V. Model Comparison

Fig. 4 provides a comparison of validation accuracies across CNN, HOG+SVM, and SIFT+SVM models. The results indicate that both CNN and HOG+SVM achieve significantly higher performance compared to SIFT+SVM. Specifically, CNN demonstrates slightly lower accuracy than HOG+SVM but offers more consistent performance across users, suggesting robust generalization to unseen signature variations. HOG+SVM, while achieving marginally higher accuracy, relies heavily on handcrafted gradient features, which may be less adaptive to diverse signature styles. In contrast, SIFT+SVM shows a substantial drop in accuracy, primarily due to the sparse and inconsistent distribution of keypoints across different signatures, resulting in incomplete or non-representative descriptors. The comparison also highlights the advantage of deep learning approaches like CNN in capturing both local and global patterns automatically, whereas traditional methods such as HOG and SIFT require careful feature engineering. Overall, the analysis confirms that end-to-end learned features with CNN provide a balanced trade-off between accuracy and robustness, while HOG remains a competitive classical method, and SIFT-based approaches are less suitable for signature verification in this dataset.

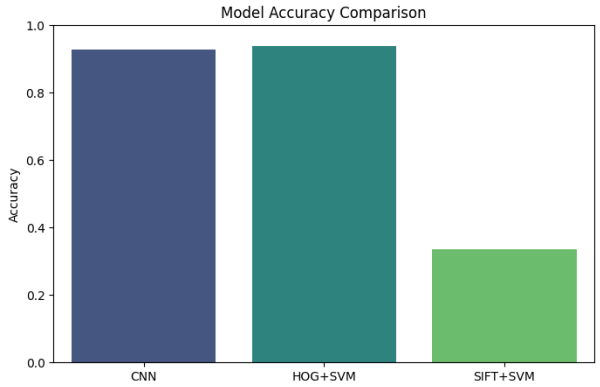

Fig. 4. CNN training and validation accuracy and loss curves over 15 epochs.

VI. Discussion

The CNN model effectively learns hierarchical features, capturing global and local patterns within signature images, leading to high classification accuracy and balanced precision recall values. HOG features remain competitive due to their robust gradient-based representation, highlighting the structural patterns of signature strokes. SIFT descriptors, while theoretically robust to transformations, are inconsistent for signature verification due to sparse keypoints and variability in descriptor counts across samples. Visualization of features demonstrates that HOG produces uniform gradient maps corresponding to signature structures, while SIFT produces sparse keypoints capturing local distinctive strokes. Overall, CNN predictions are more coherent and consistent, benefiting from end-to-end feature learning.

VII. Conclusion

CNN and HOG+SVM approaches achieve strong performance for signature verification tasks. CNN offers automated hierarchical feature learning with superior consistency, while HOG+SVM remains a viable classical alternative. SIFT+SVM underperforms due to variable keypoint density and descriptor inconsistency. Future work may explore hybrid models combining CNN with handcrafted features, or fine-tuned transfer learning approaches to further improve verification accuracy.

References

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